

PS 3: Week 5

Randomized Experiments vs. Observational Data

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Week 5

A Motivating Case: Tylenol & Autism

- Some observational data shows children whose mothers took acetaminophen during pregnancy were slightly more likely to be diagnosed with autism.
- Is this causal?

A Motivating Case: Tylenol & Autism

- Some observational data shows children whose mothers took acetaminophen during pregnancy were slightly more likely to be diagnosed with autism.
- Is this causal?
- **Omitted variable bias:** people who take Tylenol may do so because of other conditions (chronic pain, environmental factors, etc.) that could *also* raise autism risk.
- **Selection bias:** people who choose to take a “treatment” may differ, in ways connected to the outcome, from those who don’t.

A Potential Causal Effect: Populism

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- Imagine voters in a 2026 campaign: some are exposed to populist rhetoric, and each later votes for Party P or Party O.

Discuss:

- What's the **treatment** in this example?
- What are the **potential outcomes**?
- What is each individual's **treatment effect**?

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Exposure to populist rhetoric

- What are the **potential outcomes**?

Voters vote choice when exposed to populist rhetoric

Voters vote choice when not exposed to populist rhetoric

- What is each individual's **treatment effect**?

Vote when exposed to populist rhetoric - Vote when not exposed to populist rhetoric

Potential Outcomes

Voter	Vote if exposed?	Vote if not exposed?	Individual Treatment Effect
Tess	0	0	0
Jeremy	1	1	0
Anne Marie	1	0	1
Jack	0	1	-1

1: *Party P*

0: *Party O*

Observational Data on Populism

- A survey of over 1 million voters asks: were you exposed to Party P's populist rhetoric in 2026 (TV, social media, mailers, etc.)? Who did you vote for?
- People exposed to the rhetoric are **far more likely** to vote for Party P.

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- People exposed to the rhetoric are **far more likely** to vote for Party P.

Discuss:

- How confident should we be that this is *causal*?
- What sources of bias might undermine this finding?
- What are the *advantages* of a dataset like this?

Observational Data on Populism

- How confident should we be that this is *causal*?

Not confident. There is no random assignment. People could have opted in to mailers.

- What sources of bias might undermine this finding?

Selection Bias: People who remembered receiving populist rhetoric might have been more influenced by it and therefore also more likely to report voting for Party P.

Omitted Variable Bias: Underlying factors (like prior political leaning, geographic location, or media consumption) could cause both exposure to the rhetoric and the decision to vote for Party P.

Reverse Causation: Voters who already decided to vote for Party P might actively seek out their campaign materials or sympathetic media.

- What are the *advantages* of a dataset like this?

Massive sample size: 1 million voters provides high statistical power.

High external validity: It captures how real voters behave in the real world during an actual campaign, rather than in an artificial lab setting.

Omitted Variable Bias, Selection Bias, and Reverse Causation!

Voter	Exposed in 2025?	Voted P 2025?	Voted P 2021?
Tess	1	1	1
Jeremy	1	1	1
Anne Marie	0	0	0
Jack	0	0	0

How Do We Resolve These Biases?

- With observational data, people can **select into treatment**.
- If that selection *and* the outcome are both explained by some other variable, that's **omitted variable bias**.
 - E.g., men being less risk-averse → more likely to take an experimental drug *and* have lower resting heart rates.
- **Randomization** makes treatment and control groups similar on *all* variables (given a large enough sample) – no more omitted variable bias to worry about.
 - E.g., prior vote can no longer cause *both* exposure to rhetoric and future vote, because it's balanced across groups.

How might we set this up?

- What's the treatment?
- How is the treatment assigned?
- What outcome(s) are we measuring?
- What variables should we measure?
 - Often driven by which **omitted variables** we're worried about.

Designing an Experiment on Populism

- What's the treatment?

Treatment group sees a message from Party P: Party O has spent 12 years in power doing nothing for hardworking people while enriching themselves and their billionaire friends; Party P is committed to social welfare spending instead.

Control group sees nothing.

- How is the treatment assigned?

Randomly

- What outcome(s) are we measuring?

Who they'd vote for in a hypothetical P vs. O election

- What variables should we measure?

Past vote history, demographics, political knowledge, and political attention

Bundled Treatments

- To find a causal effect, you need to **isolate** your explanatory variable from your outcome.
- If your treatment actually bundles *multiple* things together, how do you know which part is doing the work?
 - Is it the **populism**? Or the **policies**?
- Solution: split the bundle across multiple treatment arms.
 - Populism treatment (previous slide)
 - Policies only, no populist framing
 - Control group
- Compare the bundled treatment to the “policies only” arm to isolate the effect of populist rhetoric *itself*.

Attendance Survey



Practice Coding Exercise 1

- Note: *This is a harder exercise than we have done in class or in the problem set before.*

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- Note: *This is a harder exercise than we have done in class or in the problem set before.*
- Using the data set on persuasion and voting from Wednesday in `Week5_Sandbox1.ipynb`
- What is the turnout percentage for each treatment? (Civic Duty, Hawthorne, Self, and Neighbors)

Answer: Coding Exercise 1

```
civic.data <- subset(social, treat_civic == 1)
hawthorne.data <- subset(social, treat_hawthorne == 1)
self.data <- subset(social, treat_self == 1)
neighbors.data <- subset(social, treat_neighbors == 1)
```

```
civic.turnout <- mean(civic.data$outcome_voted)
hawthorne.turnout <- mean(hawthorne.data$outcome_voted)
self.turnout <- mean(self.data$outcome_voted)
neighbors.turnout <- mean(neighbors.data$outcome_voted)

c(civic.turnout, hawthorne.turnout, self.turnout, neighbors.turnout)
```

```
#> [1] 0.3264837 0.3298170 0.3565500 0.3903000
```

Answer: Coding Exercise 1

We could also do it all in one step:

```
civic.turnout <- mean(subset(social, treat_civic == 1)$outcome_voted)
hawthorne.turnout <- mean(subset(social, treat_hawthorne == 1)$outcome_voted)
self.turnout <- mean(subset(social, treat_self == 1)$outcome_voted)
neighbors.turnout <- mean(subset(social, treat_neighbors == 1)$outcome_voted)

c(civic.turnout, hawthorne.turnout, self.turnout, neighbors.turnout)
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#> [1] 0.3264837 0.3298170 0.3565500 0.3903000
```

Practice Coding Exercise 2: Random Assignment

In the in-class activity we checked if age differed between the Neighbors and Control groups.

Does voter-history (g2000, g2002, p2004) differ by treatment assignment?

That is, what is the turnout rate for each election across the 4 treatments and control group?

What does this tell us about random assignment?

Practice Coding Exercise 2: Answers

Remember how we subset the data before:

```
civic.data <- subset(social, treat_civic == 1)
hawthorne.data <- subset(social, treat_hawthorne == 1)
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Why did we need to do this for this problem as well?

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neighbors.data <- subset(social, treat_neighbors == 1)
```

Why did we need to do this for this problem as well?

Because we wanted to know treatment assignment across several groups.

g2000 Random Assignment

```
table(civic.data$g2000)
```

```
#>  
#>      0      1  
#> 2698 17303
```

```
table(self.data$g2000)
```

```
#>  
#>      0      1  
#> 2753 17247
```

```
table(hawthorne.data$g2000)
```

```
#>  
#>      0      1  
#> 2658 17344
```

```
table(neighbors.data$g2000)
```

```
#>  
#>      0      1  
#> 2688 17312
```

What does this mean?

g2000 Random Assignment

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table(neighbors.data$g2000)
```

```
#>  
#>      0      1  
#> 2688 17312
```

What does this mean?

There are similar values across each condition which means random assignment of treatment likely worked

g2002 Random Assignment

```
table(civic.data$g2002)
```

```
#>  
#>      0      1  
#> 3293 16708
```

```
table(self.data$g2002)
```

```
#>  
#>      0      1  
#> 3338 16662
```

```
table(hawthorne.data$g2002)
```

```
#>  
#>      0      1  
#> 3316 16686
```

```
table(neighbors.data$g2002)
```

```
#>  
#>      0      1  
#> 3310 16690
```

p2004 Random Assignment

```
table(civic.data$p2004)
```

```
#>  
#>      0      1  
#> 11691  8310
```

```
table(self.data$p2004)
```

```
#>  
#>      0      1  
#> 11613  8387
```

```
table(hawthorne.data$p2004)
```

```
#>  
#>      0      1  
#> 11590  8412
```

```
table(neighbors.data$p2004)
```

```
#>  
#>      0      1  
#> 11528  8472
```

Acknowledgements

- Special thanks to Will Halm and prior GSIs for providing materials off of which some of these slides are based.